26. ABAQUS - Modelling

1.) LARGE :

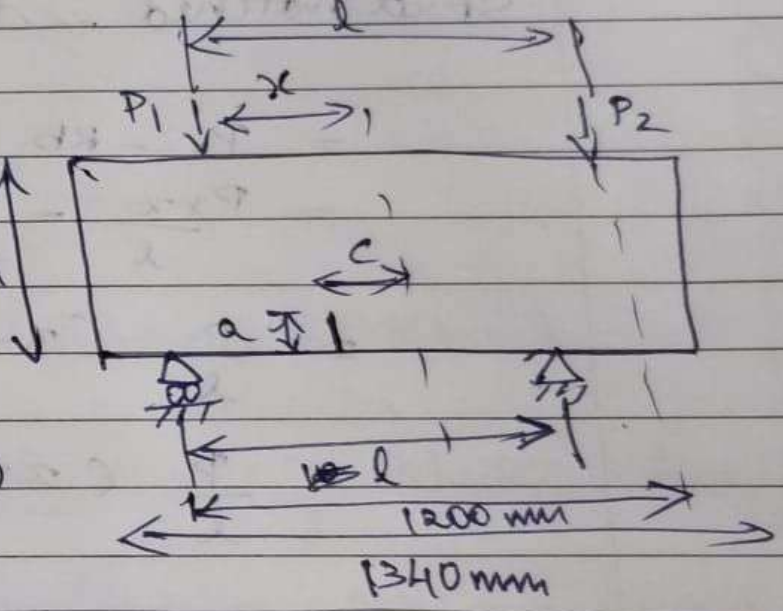
$l = 1000$ mm

$x = 400$ mm

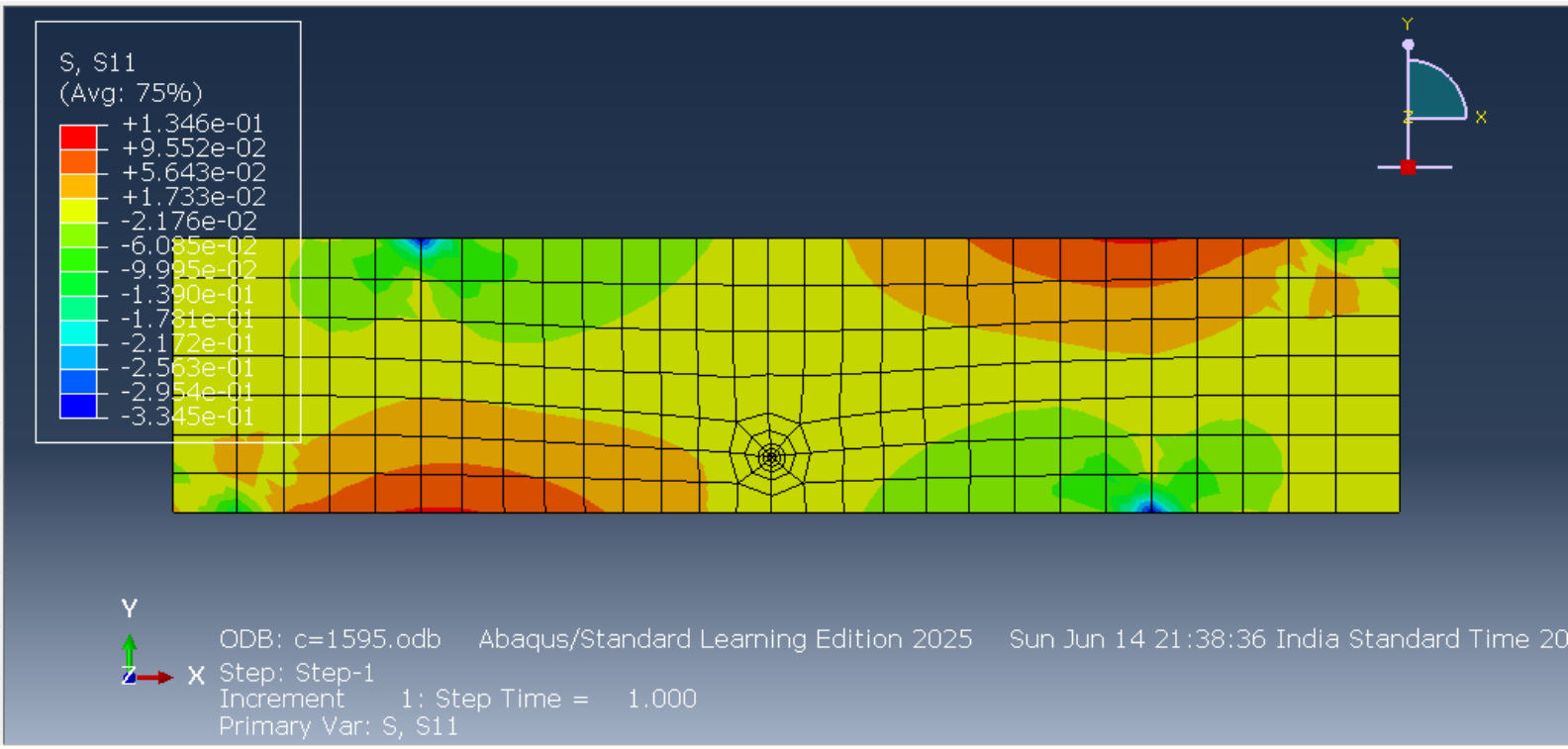
$a = 60$ mm.

$P = 1000N$

($P_1 = 600N$, $P_2 = 400N$)

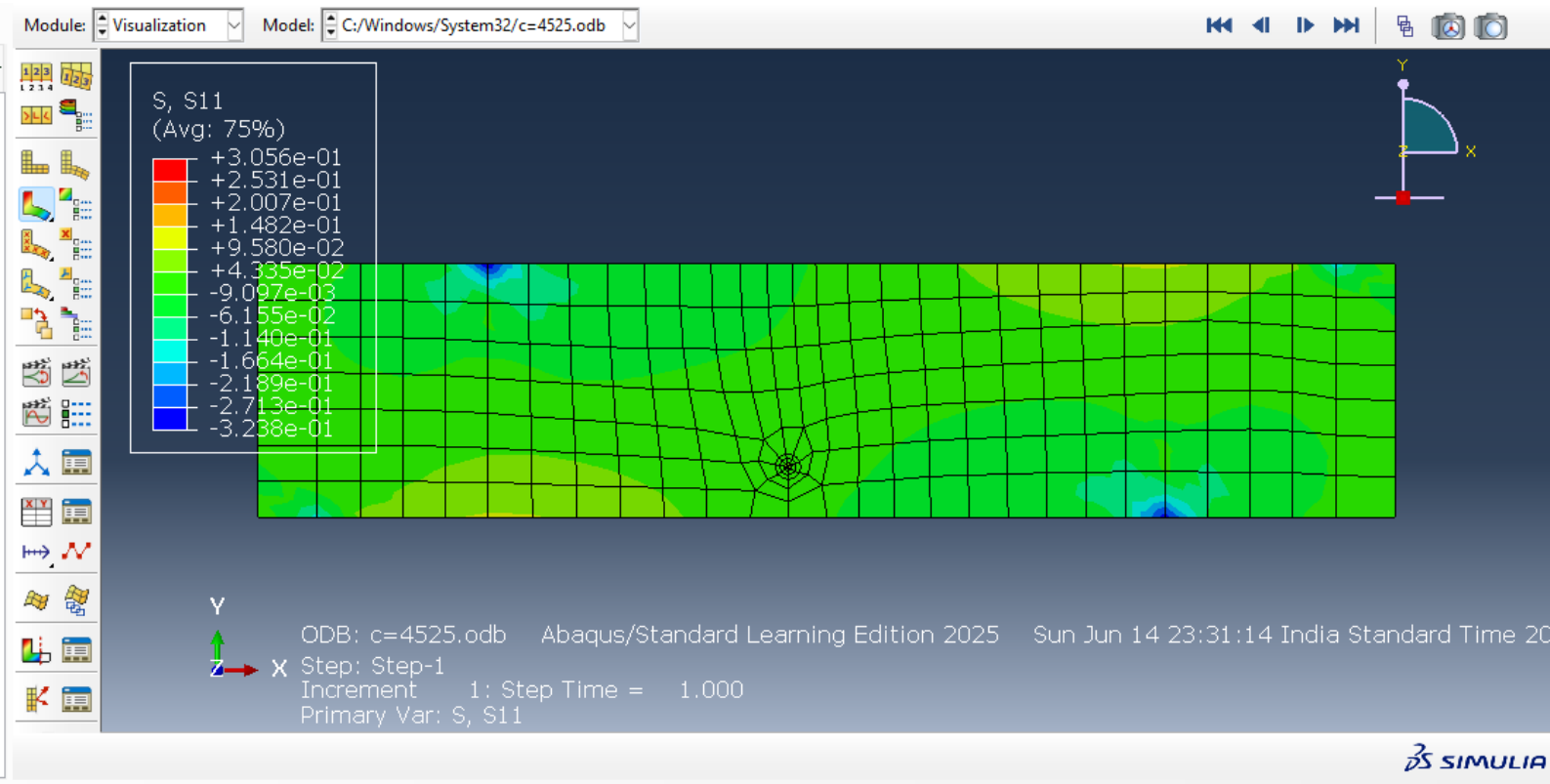


	A	B	C	D	E	F	I	J	M	N	O	P	Q	R	S	T	U
1																	
2							ABAQUS MODEL - 0.3										
3																	
4																	
5	l	x	c	a	Ra	Rb	P1	P2	K1	K2	Me	b	d	Z	bd	Load(p)	
6	1000	400	15.75	60	400	600	600	400	0.06481	0.1284	0.29759	50	300	750000	15000	1000	
7	1000	400	15.8	60	400	600	600	400	0.0619	0.1284	0.28598	50	300	750000	15000	1000	
8	1000	400	15.9	60	400	600	600	400	0.06539	0.1284	0.29988	50	300	750000	15000	1000	
9	1000	400	15.95	60	400	600	600	400	0.06557	0.1284	0.30057	50	300	750000	15000	1000	FINAL
10	1000	400	16	60	400	600	600	400	0.06574	0.1284	0.30126	50	300	750000	15000	1000	
11																	
12																	


S, S11
(Avg: 75%)
+1.346e-01
+9.552e-02
+5.643e-02
+1.733e-02
-2.176e-02
-6.085e-02
-9.995e-02
-1.390e-01
-1.781e-01
-2.172e-01
-2.563e-01
-2.954e-01
-3.345e-01
Y
X
Y
Z
X
ODB: c=1595.odb Abaqus/Standard Learning Edition 2025 Sun Jun 14 21:38:36 India Standard Time 20
Step: Step-1
Increment 1: Step Time = 1.000
Primary Var: S, S11

ABAQUS MODEL - 0.6

l	x	c	a	Ra	Rb	P1	P2	K1	K2	Me	b	d	Z	bd	Load(p)	
1000	400	60	60	400	600	600	400	0.2386	0.1315	0.67933	50	300	750000	15000	1000	
1000	400	58	60	400	600	600	400	0.2295	0.1299	0.67211	50	300	750000	15000	1000	
1000	400	50	60	400	600	600	400	0.199	0.1307	0.63004	50	300	750000	15000	1000	
1000	400	44	60	400	600	600	400	0.1753	0.1305	0.59261	50	300	750000	15000	1000	
1000	400	45	60	400	600	600	400	0.1793	0.1304	0.5997	50	300	750000	15000	1000	
1000	400	45.25	61	400	600	600	400	0.1802	0.1306	0.60075	50	300	750000	15000	1000	FINAL


Module: Visualization
Model: C:/Windows/System32/c=4525.odb
S, S11
(Avg: 75%)
+3.056e-01
+2.531e-01
+2.007e-01
+1.482e-01
+9.580e-02
+4.335e-02
-9.097e-03
-6.155e-02
-1.140e-01
-1.664e-01
-2.189e-01
-2.713e-01
-3.238e-01
Y
X
Y
Z
X
ODB: c=4525.odb Abaqus/Standard Learning Edition 2025 Sun Jun 14 23:31:14 India Standard Time 20
Step: Step-1
Increment 1: Step Time = 1.000
Primary Var: S, S11
SIMULIA

9.) MEDIUM

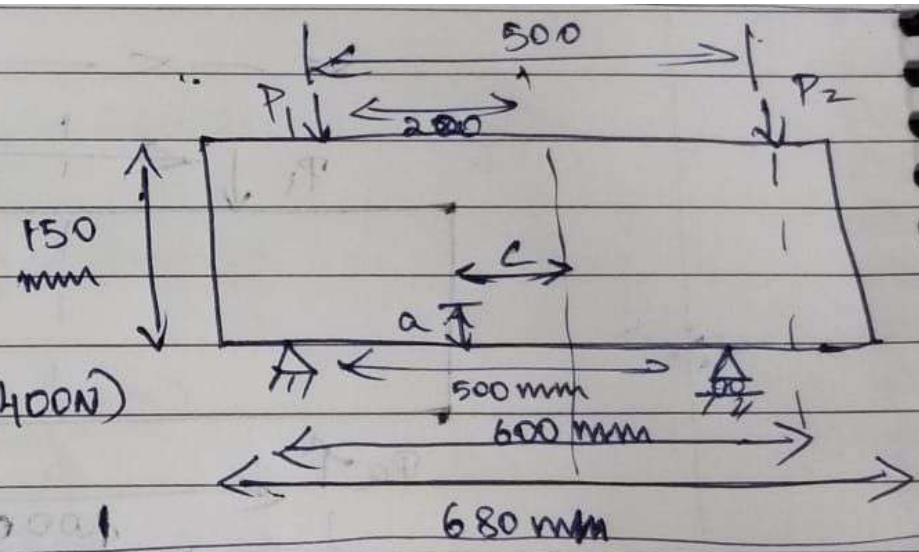
$l = 500\ mm$

$x = 200\ mm$

$a = 30\ mm$

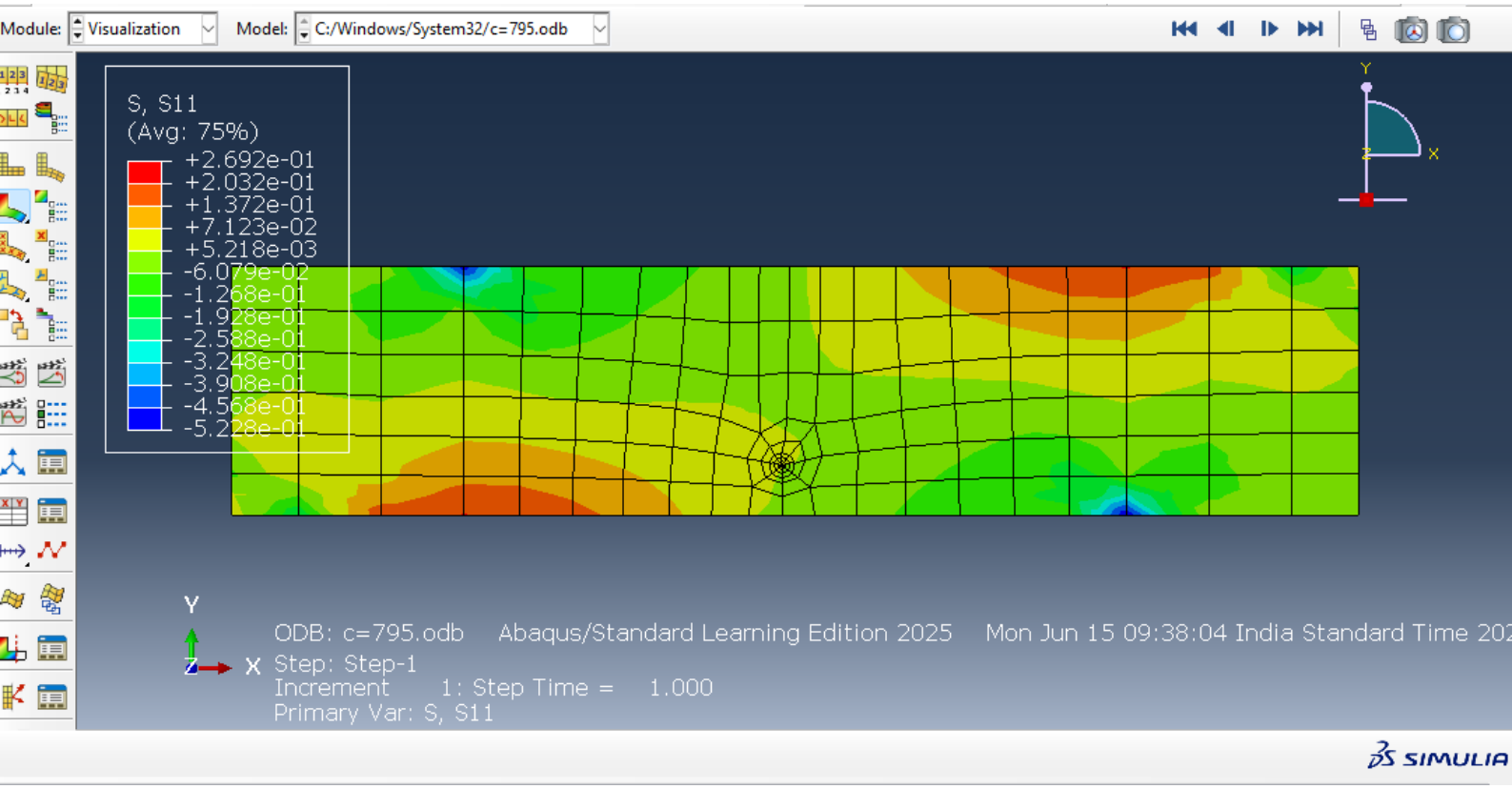
$P = 1000N$

$(P_1 = 600N,\ P_2 = 400N)$

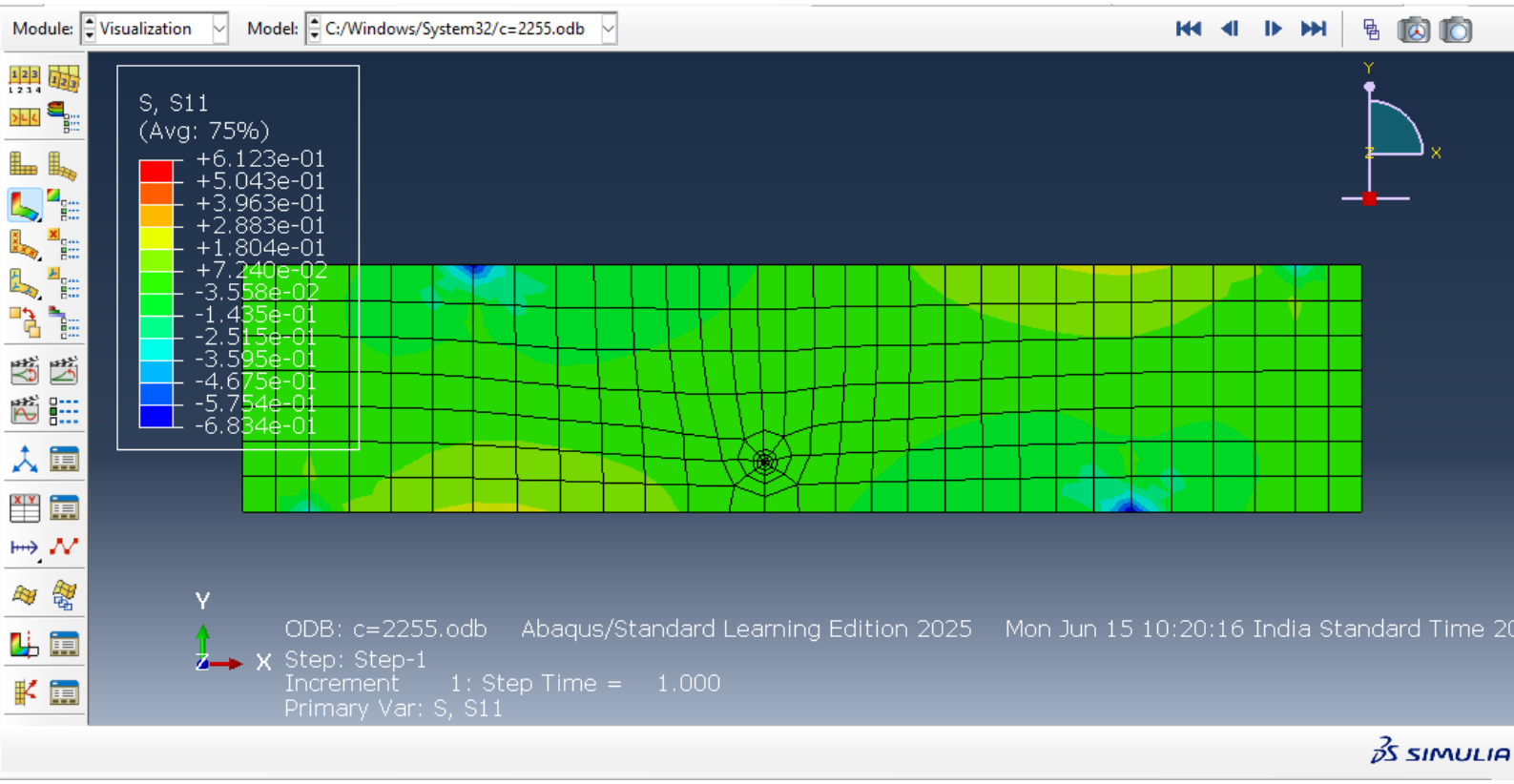


ABAQUS MODEL - 0.3

l	x	c	a	Ra	Rb	P1	P2	K1	K2	Me	b	d	Z	bd	Load(p)	
500	200	7.5	30	400	600	600	400	0.08717	0.1814	0.28518	50	150	187500	7500	1000	
500	200	8	30	400	600	600	400	0.09304	0.1816	0.30143	50	150	187500	7500	1000	
500	200	7.75	30	400	600	600	400	0.09032	0.1813	0.29424	50	150	187500	7500	1000	
500	200	7.95	30	400	600	600	400	0.09249	0.1815	0.30004	50	150	187500	7500	1000	FINAL


Module: Visualization
Model: C:/Windows/System32/c=795.odb
S, S11
(Avg: 75%)
+2.692e-01
+2.032e-01
+1.372e-01
+7.123e-02
+5.218e-03
-6.079e-02
-1.268e-01
-1.928e-01
-2.588e-01
-3.248e-01
-3.908e-01
-4.568e-01
-5.228e-01
Y
X
Z
ODB: c=795.odb Abaqus/Standard Learning Edition 2025 Mon Jun 15 09:38:04 India Standard Time 202
Step: Step-1
Increment 1: Step Time = 1.000
Primary Var: S, S11
SIMULIA

	A	B	C	D	E	F	I	J	M	N	O	P	Q	R	S	T	U
1																	
2						ABAQUS MODEL - 0.6											
3																	
4	l	x	c	a	Ra	Rb	P1	P2	K1	K2	Me	b	d	Z	bd	Load(p)	
5	500	200	20	30	400	600	600	400	0.2255	0.1837	0.56481	50	150	187500	7500	1000	
6	500	200	22	30	400	600	600	400	0.2479	0.184	0.59351	50	150	187500	7500	1000	
7	500	200	22.5	30	400	600	600	400	0.2535	0.1845	0.59947	50	150	187500	7500	1000	
8	500	200	22.75	30	400	600	600	400	0.2548	0.183	0.60349	50	150	187500	7500	1000	
9	500	200	22.6	30	400	600	600	400	0.2531	0.183	0.60146	50	150	187500	7500	1000	
10	500	200	22.55	30	400	600	600	400	0.254	0.1845	0.60007	50	150	187500	7500	1000	FINAL
11																	
12																	


Module: Visualization
Model: C:/Windows/System32/c=2255.odb
S, S11
(Avg: 75%)
+6.123e-01
+5.043e-01
+3.963e-01
+2.883e-01
+1.804e-01
+7.240e-02
-3.558e-02
-1.435e-01
-2.515e-01
-3.595e-01
-4.675e-01
-5.754e-01
-6.834e-01
Y
X
Z
ODB: c=2255.odb Abaqus/Standard Learning Edition 2025 Mon Jun 15 10:20:16 India Standard Time 20
Step: Step-1
Increment 1: Step Time = 1.000
Primary Var: S, S11
SIMULIA

3.) SMALL

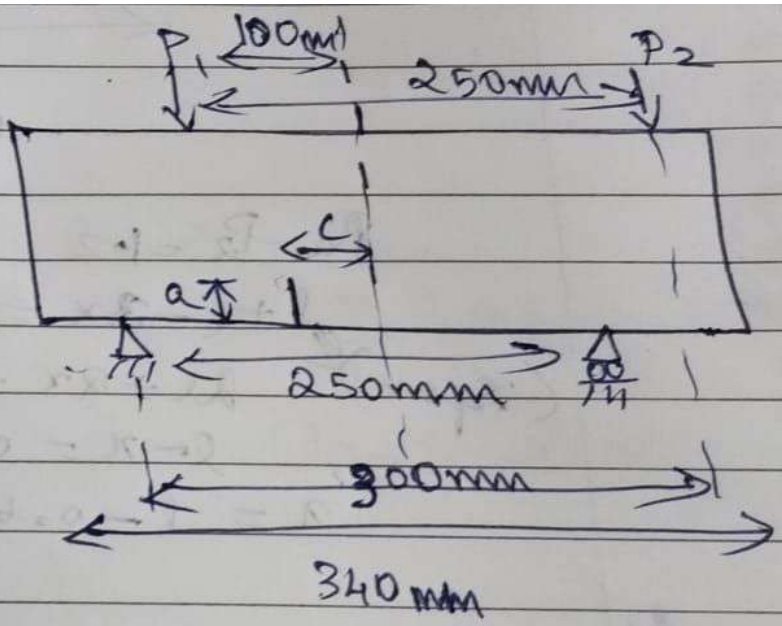
$l = 250\,mm$

$x = 100\,mm$

$a = 15\,mm$

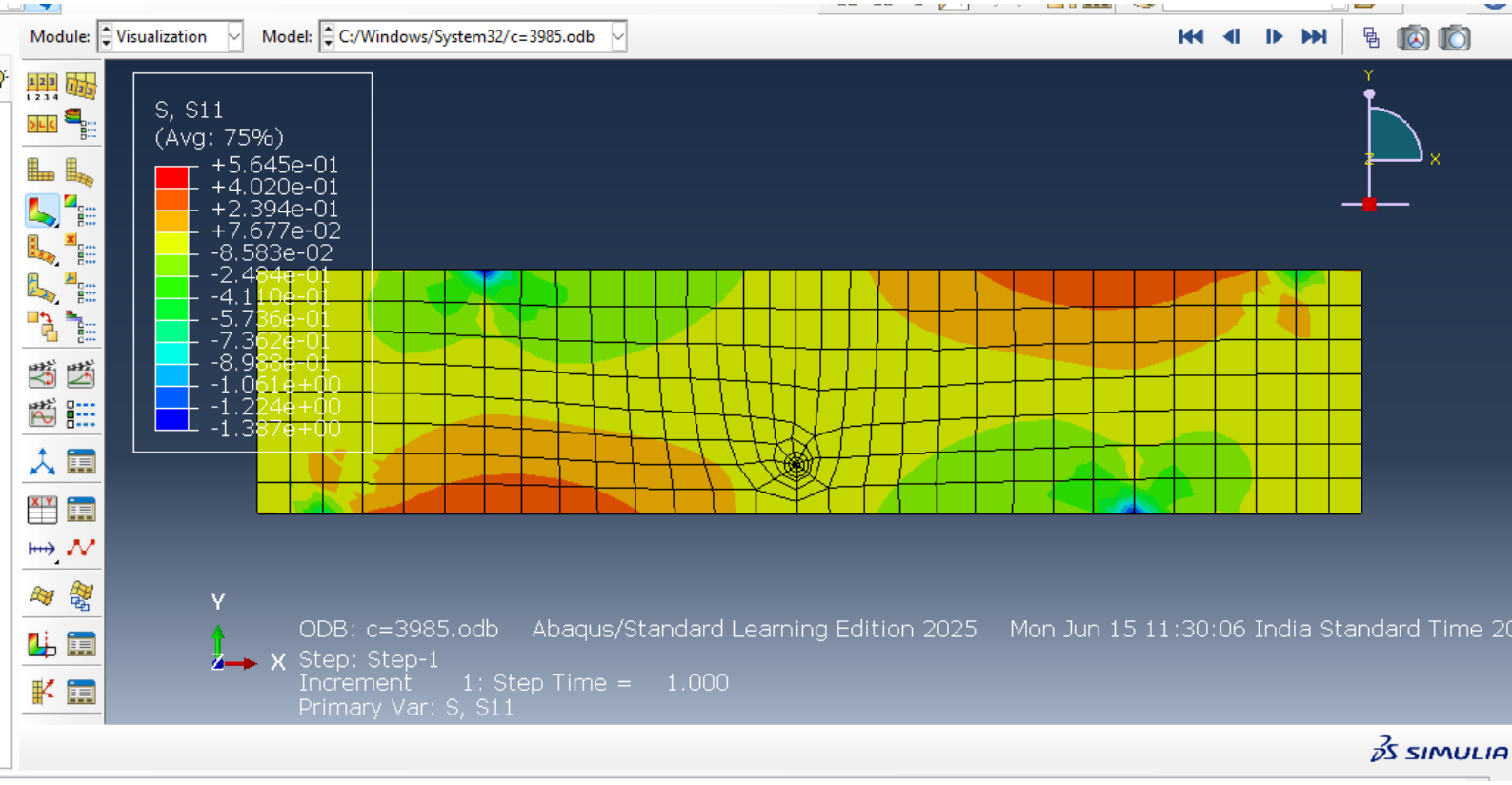
$P = 1000\,N$

$(P_1 = 600\,N,\ P_2 = 400\,N)$

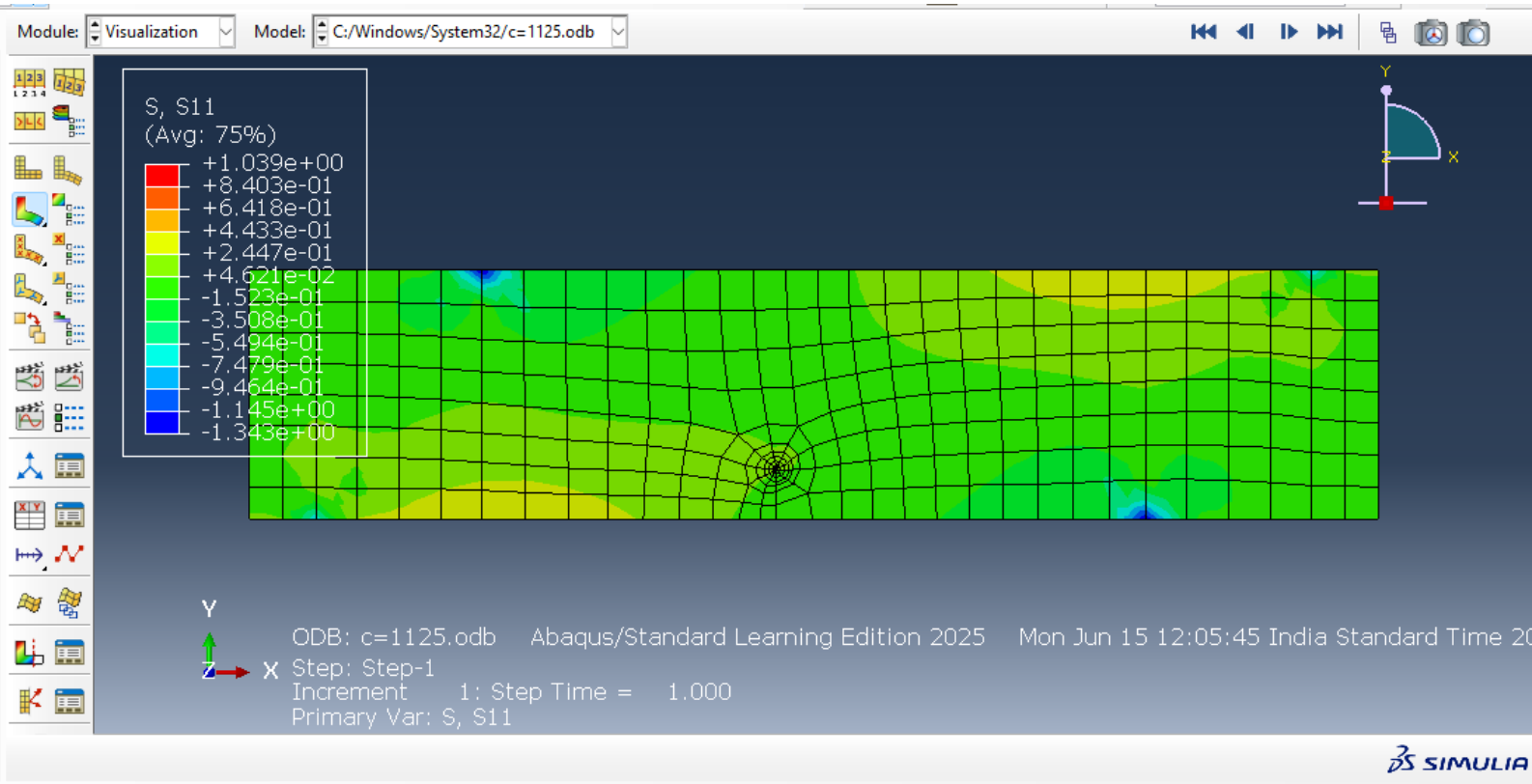


ABAQUS MODEL - 0.3

l	x	c	a	Ra	Rb	P1	P2	K1	K2	Me	b	d	Z	bd	Load(p)	
250	100	4	15	400	600	600	400	0.1313	0.2562	0.3015	50	75	46875	3750	1000	
250	100	3.8	15	400	600	600	400	0.1249	0.2564	0.28858	50	75	46875	3750	1000	
250	100	3.95	15	400	600	600	400	0.1294	0.2561	0.29785	50	75	46875	3750	1000	
250	100	3.98	15	400	600	600	400	0.1303	0.2562	0.29953	50	75	46875	3750	1000	
250	100	3.985	15	400	600	600	400	0.1314	0.2573	0.30059	50	75	46875	3750	1000	FINAL


Module: Visualization
Model: C:/Windows/System32/c=3985.odb
S, S11
(Avg: 75%)
+5.645e-01
+4.020e-01
+2.394e-01
+7.677e-02
-8.583e-02
-2.484e-01
-4.110e-01
-5.736e-01
-7.362e-01
-8.988e-01
-1.061e+00
-1.224e+00
-1.387e+00
Y
X
Z
ODB: c=3985.odb Abaqus/Standard Learning Edition 2025 Mon Jun 15 11:30:06 India Standard Time 20
Step: Step-1
Increment 1: Step Time = 1.000
Primary Var: S, S11
SIMULIA

	A	B	C	D	E	F	I	J	M	N	O	P	Q	R	S	T	U
1																	
2						ABAQUS MODEL - 0.6											
3																	
4	l	x	c	a	Ra	Rb	P1	P2	K1	K2	Me	b	d	Z	bd	Load(p)	
5	250	100	12	15	400	600	600	400	0.3818	0.2608	0.61849	50	75	46875	3750	1000	
6	250	100	11.5	15	400	600	600	400	0.3649	0.2596	0.60634	50	75	46875	3750	1000	
7	250	100	11	15	400	600	600	400	0.351	0.2623	0.59144	50	75	46875	3750	1000	
8	250	100	11.25	15	400	600	600	400	0.3587	0.2605	0.60013	50	75	46875	3750	1000	FINAL
9																	
10																	


Module: Visualization
Model: C:/Windows/System32/c=1125.odb
S, S11
(Avg: 75%)
+1.039e+00
+8.403e-01
+6.418e-01
+4.433e-01
+2.447e-01
+4.621e-02
-1.523e-01
-3.508e-01
-5.494e-01
-7.479e-01
-9.464e-01
-1.145e+00
-1.343e+00
Y
X
Z
ODB: c=1125.odb Abaqus/Standard Learning Edition 2025 Mon Jun 15 12:05:45 India Standard Time 20
Step: Step-1
Increment 1: Step Time = 1.000
Primary Var: S, S11
SIMULIA